

CS23334 – FUNDAMENTALS OF DATA SCIENCE

1(A) - Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib /seaborn for visualization.

```
import pandas as pd

import matplotlib.pyplot as plt

data = {
    'Year': list(range(2010, 2021)),
    'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]
}

df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

plt.ylabel('Number of Job Postings')

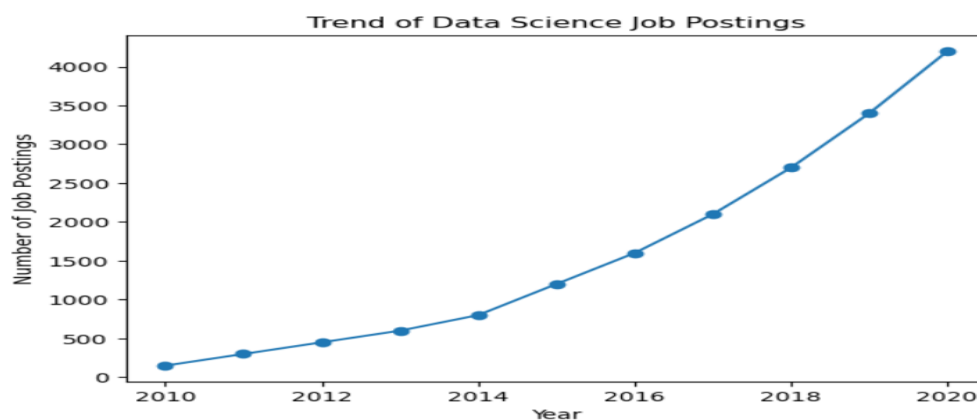
plt.show()
```

Sample Data Input:

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Output:



Exp No:1.b Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

```
import matplotlib.pyplot as plt
```

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer', 'Business Analyst']
```

```
counts = [300, 500, 450, 200, 150]
```

```
plt.bar(roles, counts)
```

```
plt.title('Distribution of Data Science Roles')
```

```
plt.xlabel('Role')
```

```
plt.ylabel('Count')
```

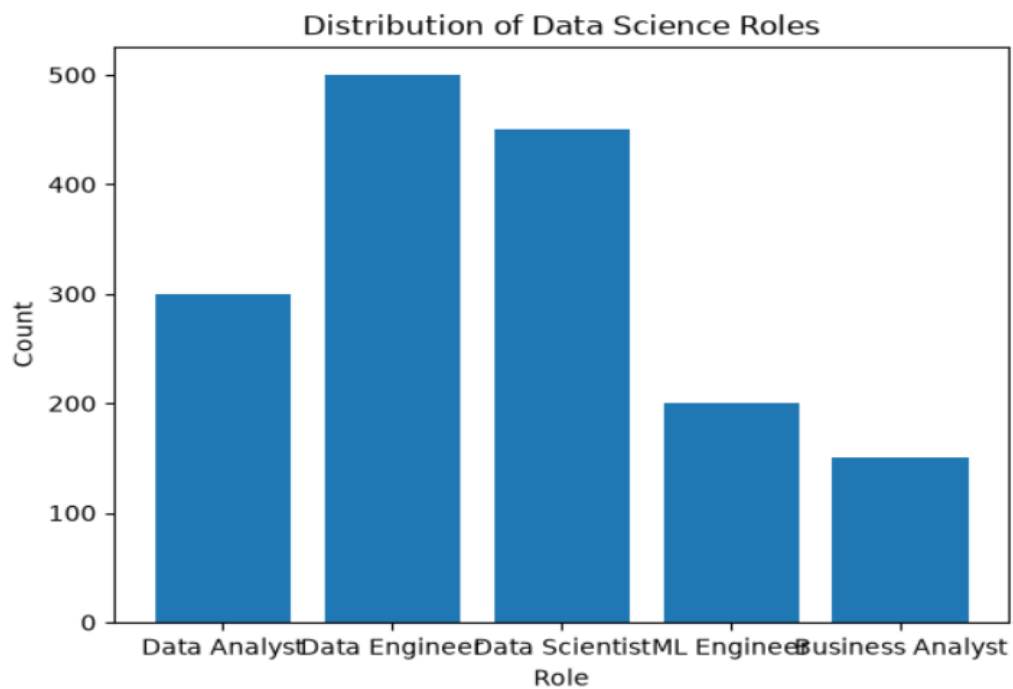
```
plt.show()
```

Sample Input:

```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer', 'Business Analyst']
```

```
counts = [300, 500, 450, 200, 150]
```

Sample Output:



Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

```
import pandas as pd

structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})

print("Structured Data:\n", structured_data)

unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image, or a video file."

print("\nUnstructured Data:\n", unstructured_data)

semi_structured_data = {
    'ID': 1,
    'Name': 'Alice',
    'Attributes': {
        'Height': 165,
        'Weight': 68
    }
}

print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

OUTPUT:

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Charlie	35

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

```
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
}
```

Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data
from cryptography.fernet import Fernet

```
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhylZCl-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn
XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWdf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8FovltV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```